

# User Flows – Current vs Shamir Recovery

Side-by-side comparison of the current new-wallet and seed-import flows alongside the proposed Shamir DEK sharding model. Onboarding is unchanged; recovery is post-onboard.

CURRENTNEW WALLET

ONBOARDING

1

Welcome → "Get started"

2

PIN setup (8-12 digits) + confirm

3

Explore mode (empty dashboard)

4

Tap "Create new wallet"

5

Provisioning  
seed → Argon2id → AES-256-GCM → vault

6

KEK auto-enroll  
biometric → H → HKDF(H||C) → wrap DEK

7

Telemetry consent

8

FirstRunTour (5-step modal)

Dashboard ready

8 steps · ~60-90 sec

RECOVERY

-

Manual backup envelope export only  
Most users never export

CURRENTSEED IMPORT

ONBOARDING

1

Welcome → "Get started"

2

PIN setup (8-12 digits) + confirm

3

Explore mode (empty dashboard)

4

"Import seed phrase"  
Enter 12 or 24 words

- 5

Provisioning

validate → Argon2id → AES-256-GCM → vault
- 6

KEK auto-enroll

biometric → H → HKDF(H||C) → wrap DEK
- 7

Telemetry consent

Dashboard ready

7 steps · ~60-90 sec

No FirstRunTour on import

RECOVERY

- Manual backup envelope export only

Most users never export

PROPOSED

SHAMIR MODEL

ONBOARDING (UNCHANGED)

- 1

Welcome → "Get started"
- 2

PIN setup (8-12 digits) + confirm
- 3

Explore mode (empty dashboard)
- 4

Create / Import wallet
- 5

Provisioning

seed → Argon2id → AES-256-GCM → vault
- 6

KEK auto-enroll

HKDF(H||C) – zero changes
- 7

Consent + Tour

Dashboard ready – ~48% posture

7-8 steps · identical to current

POST-ONBOARD RECOVERY (FROM POSTURE)

- R1

PIN + biometric re-auth
- R2

Set recovery passphrase

Min 16 chars · Argon2id → Recovery Key
- R3

Shamir split DEK (2-of-3)

$GF(2^8)$  · `crypto.getRandomValues`
- R4

Display Share C as QR
- R5

Scan to verify

Constant-time compare
- R6

Persist shares

A → KEK-wrap · B → RK-wrap → cloud · cache DEK

**Recovery active – posture ≥91%**

+30–45 sec when user is ready

Onboarding is **identical** across all three flows. Recovery setup is triggered by the **Security Posture Assessment** on the dashboard – not during onboarding.

**KEY INVARIANTS**

- KEK = HKDF(H||C) – zero changes in all flows
- Vault AES-256-GCM encryption – unchanged
- Shamir splits the DEK after generation, before KEK wrapping
- Share A is 32 bytes – identical to current full DEK from KEK's perspective
- Fast-path DEK cache: normal unlock has zero additional latency
- Deniability: each credential set gets independent shard sets
- Recovery setup is always skippable – 3-skip cooldown, then banner only